PocketMate CM5 carrier v0.1 — see hardware/block-diagram.md for block diagram, power tree and power states

CM5

File: cm5.kicad_sch

DisplayCamera

File: display_camera.kicad_sch

Audio

File: audio.kicad_sch

PowerInput

File: power_input.kicad_sch

Regulators

File: regulators.kicad_sch

MCU

File: mcu.kicad_sch

Sensors

File: sensors_haptics.kicad_sch

SideFlex

File: side_flex.kicad_sch

LTE

File: lte_link.kicad_sch

PocketMate

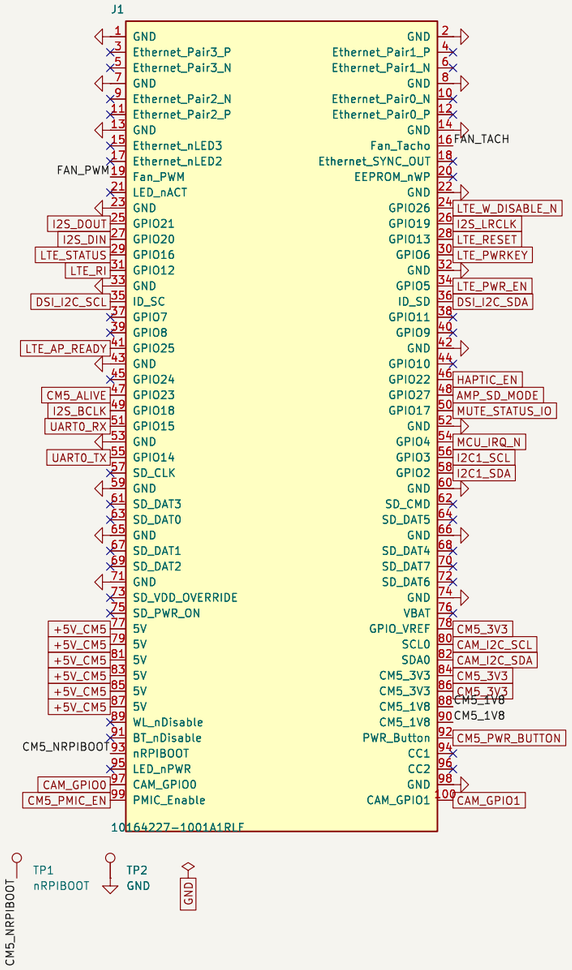
Sheet: /

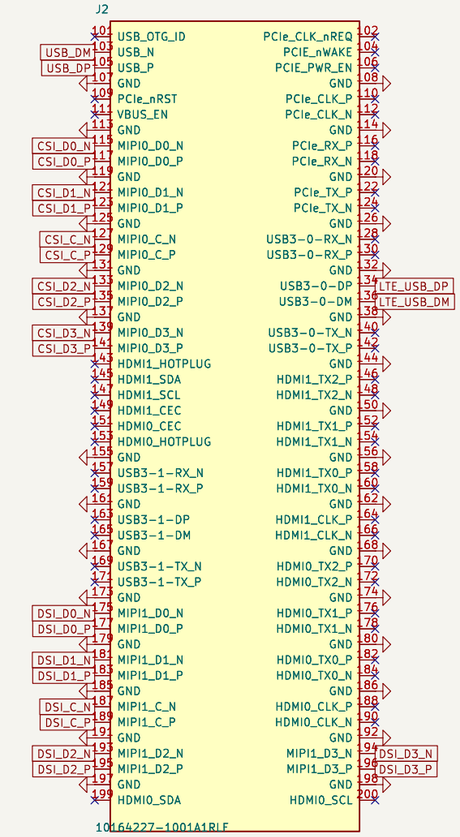
File: PocketMate.kicad_sch

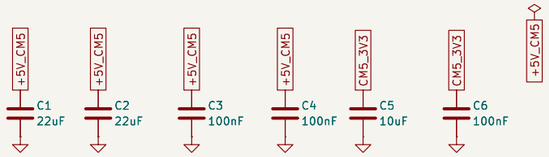
Title: PocketMate CM5 carrier — top level

Size: A3	Date: 2026-10-21	**Rev: 0.1**
KiCad E.D.A. 9.0.2		Id: 1/10

CM5 module B2B connectors. Connections are by net label at pin endpoints; global labels cross sheets.





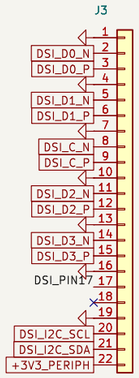


CM5 module B2B connectors
PocketMate
Sheet: /CM5/
File: cm5.kicad_sch

Title: PocketMate – CM5

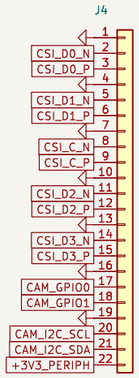
Size: A3	Date: 2026-10-21	**Rev: 0.1**
KiCad E.D.A. 9.0.2		Id: 2/10

DSI display (Waveshare 4.3in) and CSI camera (IMX708 / Camera Module 3). Connections are by net label at pin endpoints; global labels cross sheets.

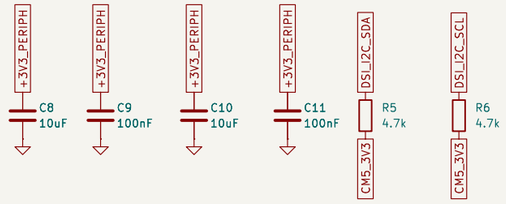


DSI1 22p 0.5mm FH12-22S-0.5SH (Pi5 DSI pinout)





CSI0 22p 0.5mm FH12-22S-0.5SH (Pi5 CSI pinout)



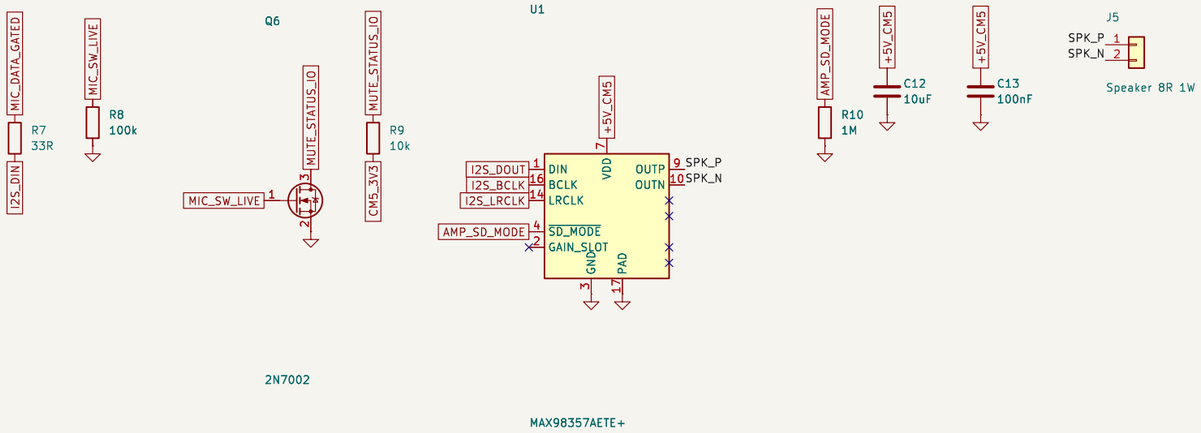
DSI display (Waveshare 4.3in) and CSI camera (IMX708 / Camera Module 3)
PocketMate

Sheet: /DisplayCamera/
File: display_camera.kicad_sch

Title: PocketMate – DisplayCamera

Size: A3	Date: 2026-10-21	**Rev: 0.1**
KiCad E.D.A. 9.0.2		Id: 3/10

MAX98357A amp; mic data/mute-status interface to the side-key flex. Connections are by net label at pin endpoints; global labels cross sheets.



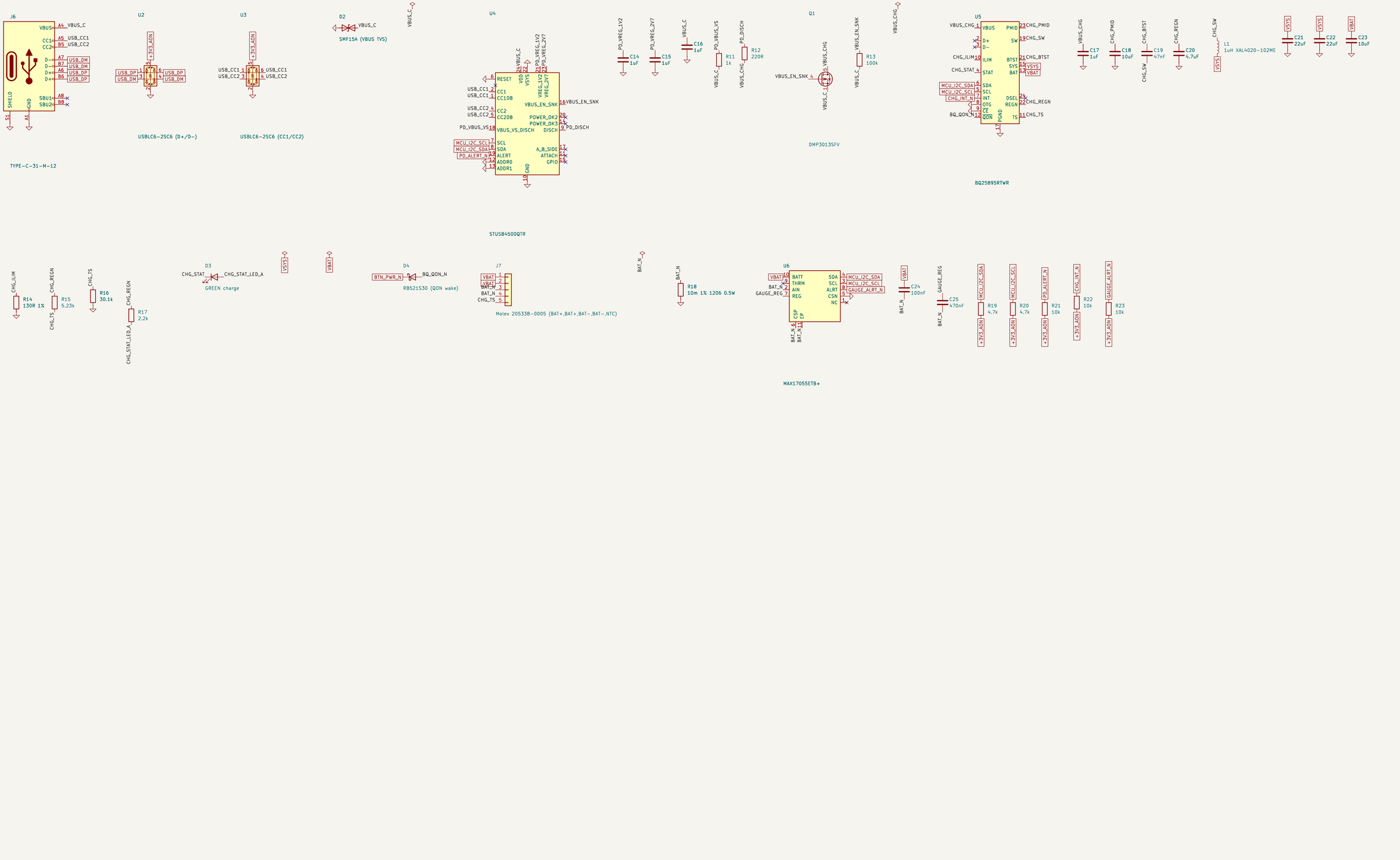
MAX98357A amp; mic data/mute-status interface to the side-key flex
PocketMate

Sheet: /Audio/
File: audio.kicad_sch

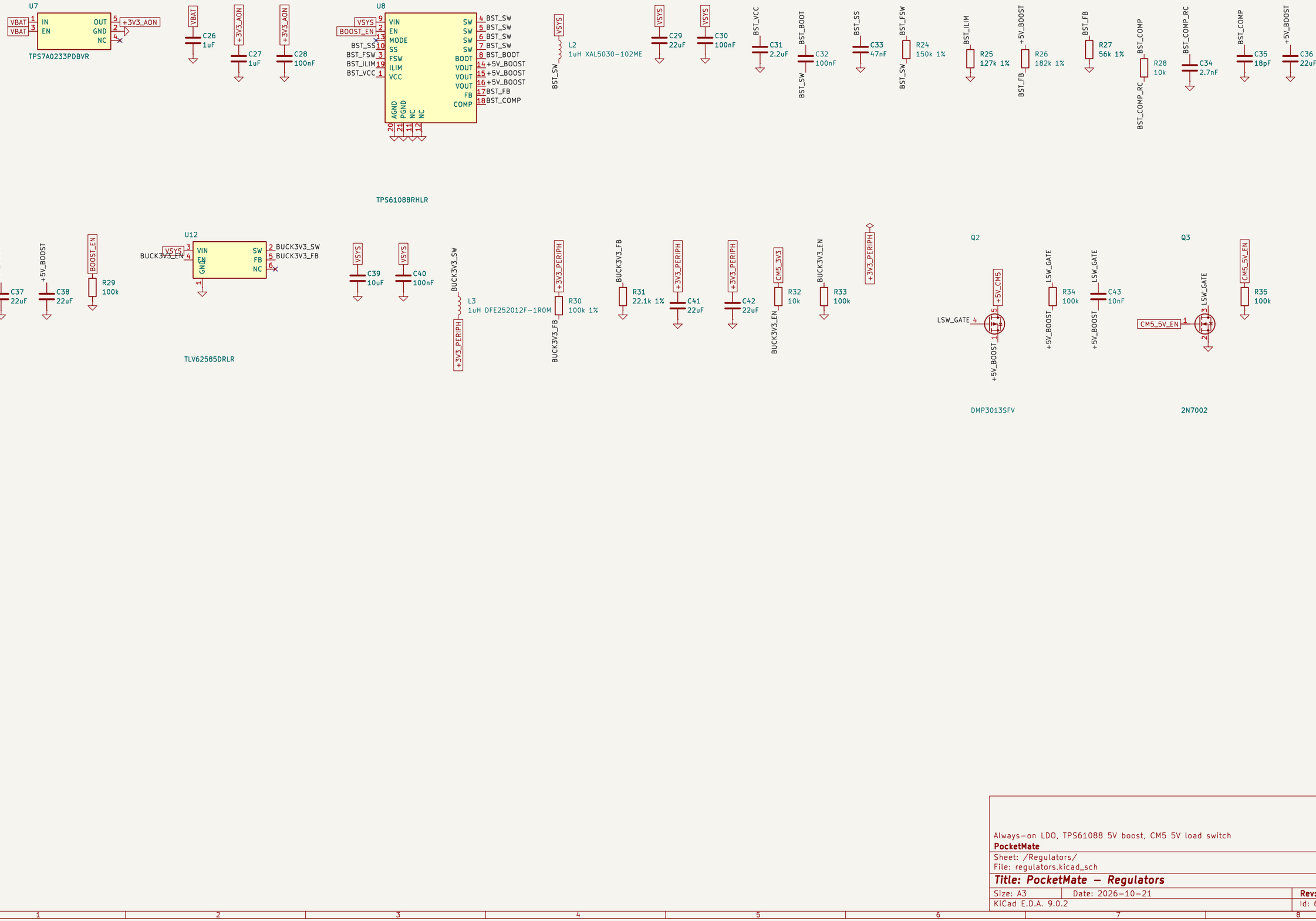
Title: PocketMate — Audio

Size: A3	Date: 2026-10-21	**Rev: 0.1**
KiCad E.D.A. 9.0.2		Id: 4/10

USB-C, STUSB4500 PD sink, BQ25895 charger, MAX17055 gauge, battery. Connections are by net label at pin endpoints; global labels cross sheets.



Always-on LDO, TPS61088 5V boost, CM5 5V load switch. Connections are by net label at pin endpoints; global labels cross sheets.



Always-on LDO, TPS61088 5V boost, CM5 5V load switch
PocketMate

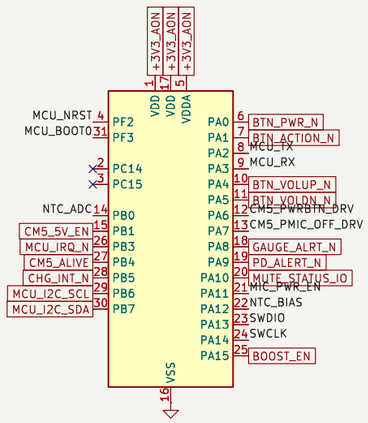
Sheet: /Regulators/
File: regulators.kicad_sch

Title: PocketMate – Regulators

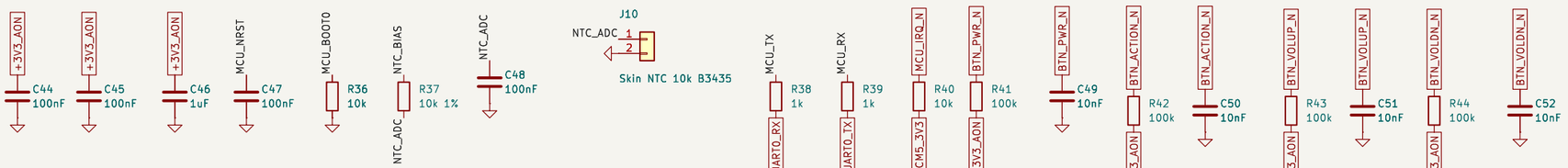
Size: A3	Date: 2026-10-21	**Rev: 0.1**
KiCad E.D.A. 9.0.2		Id: 6/10

STM32U031 always-on supervisor, buttons, CM5 power control. Connections are by net label at pin endpoints; global labels cross sheets.

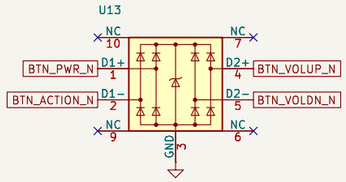
U9



STM32U031K8U6



U13



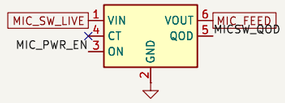
TPD4E05U06DQAR (flex buttons ESD)

D5

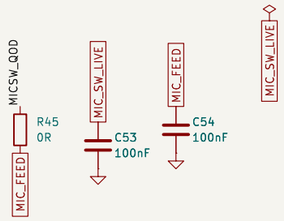


PESD5V0S1BB (ESD, MIC_SW_LIVE)

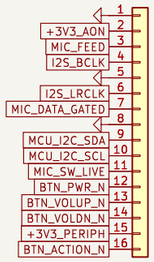
U14



TPS22917DBVR

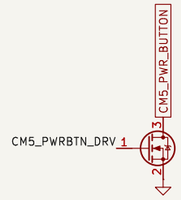


J11



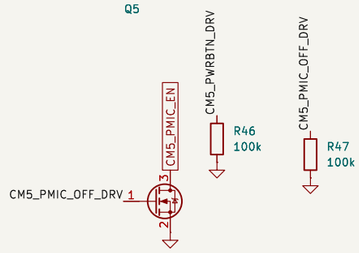
FH12-16S-0.5SH(55)

Q4



2N7002

Q5



2N7002

J8



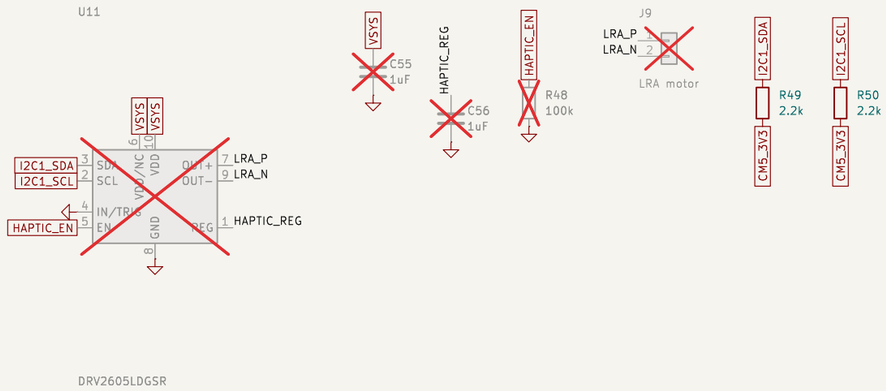
SWD (TC2030 pads)

STM32U031 always-on supervisor, buttons, CM5 power control
PocketMate
Sheet: /MCU/
File: mcu.kicad_sch

Title: PocketMate – MCU

Size: A3	Date: 2026-10-21	**Rev: 0.1**
KiCad E.D.A. 9.0.2		Id: 7/10

DRV2605L haptic (DNP), I2C1 pull-ups (VEML7700 is on the side-key flex). Connections are by net label at pin endpoints; global labels cross sheets.



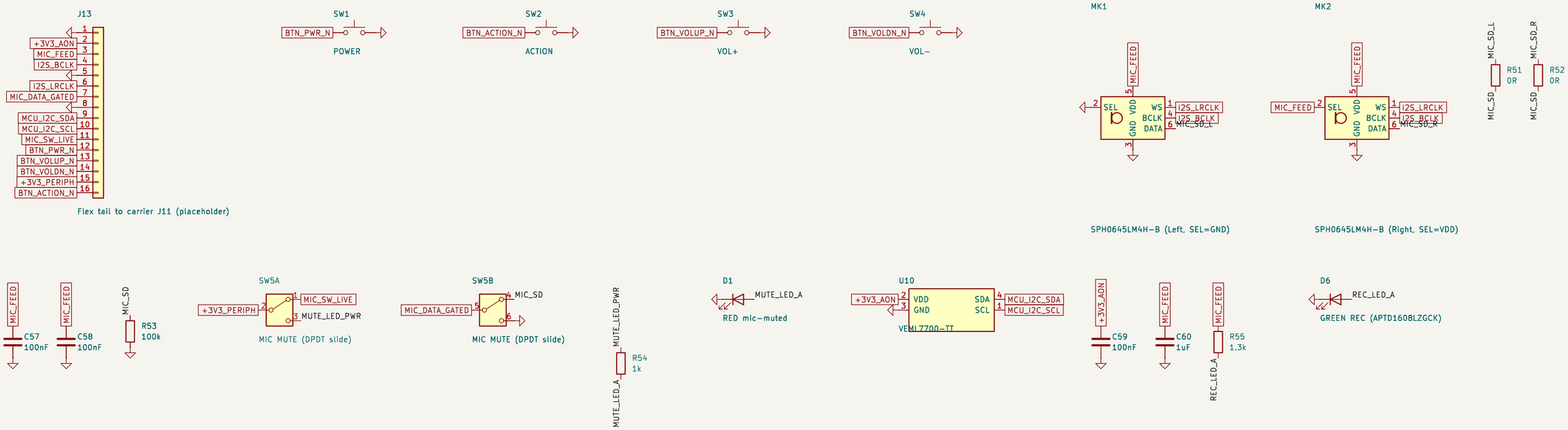
DRV2605L haptic (DNP), I2C1 pull-ups (VEML7700 is on the side-key flex)
PocketMate

Sheet: /Sensors/
File: sensors_haptics.kicad_sch

Title: PocketMate – Sensors

Size: A3	Date: 2026-10-21	**Rev: 0.1**
KiCad E.D.A. 9.0.2		Id: 8/10

Side-key flex: Power/Vol buttons, Action, hardware mic mute (DPDT + red LED), 2x SPH0645 mics, VEML7700. Connections are by net label at pin endpoints; global labels cross sheets.



Side-key flex: Power/Vol buttons, Action, hardware mic mute (DPDT + red LED), 2x SPH0645 mics, VE

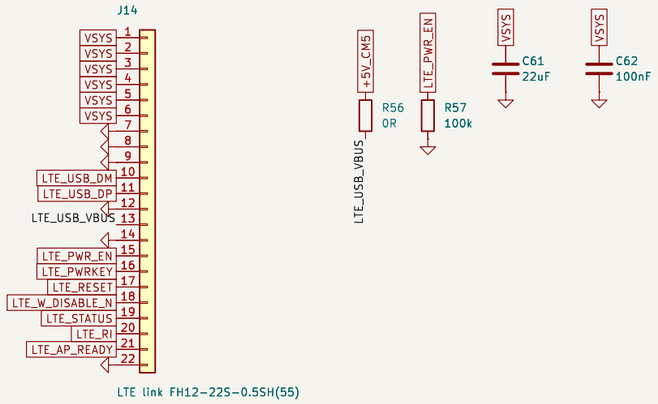
PocketMate

Sheet: /SideFlex/
File: side_flex.kicad_sch

Title: PocketMate — SideFlex

Size: A3	Date: 2026-10-21	**Rev: 0.1**
KiCad E.D.A. 9.0.2		Id: 9/10

LTE daughterboard link J14 (FH12-22S, 22p 0.5mm): VSYS, USB2 (CM5 USB3-0 DP/DM), control GPIOs. Connections are by net label at pin endpoints; global labels cross sheets.



LTE daughterboard link J14 (FH12-22S, 22p 0.5mm): VSYS, USB2 (CM5 USB3-0 DP/DM), control GPIO

PocketMate

Sheet: /LTE/
File: lte_link.kicad_sch

Title: PocketMate – LTE

Size: A3	Date: 2026-10-21	**Rev: 0.1**
KiCad E.D.A. 9.0.2		Id: 10/10